GEK-40377A

DRIVE SYSTEM INSTRUCTIONS

TURBINE CONTROL ELEMENTARY READING AID

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

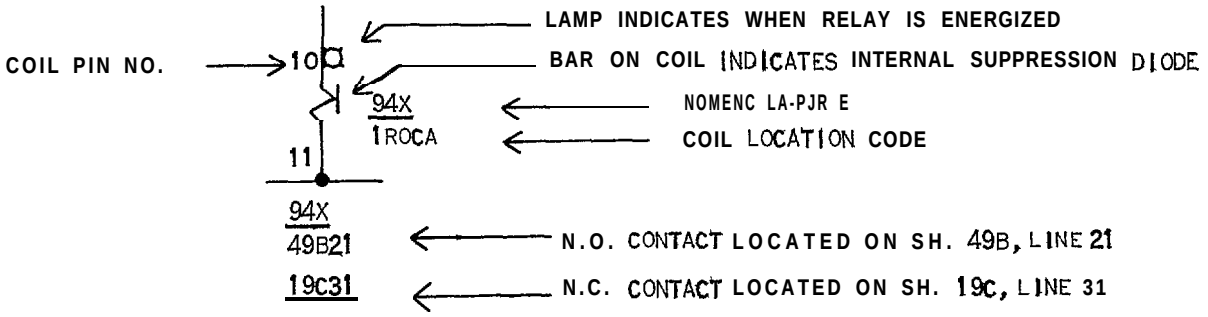
GENERAL ELECTRIC

CONTENTS

		Page
I.	Notes on Reading the Elementary Diagram	3
	A. Coil Nomenclature, Coil Location Code and Contact Locations	3
	B. Wire Reference	4
	C. Potentiometers	4
	D. Terminal Boards and Connectors	4
	1. Terminal Boards	4
	2. Plug/Receptacle Type Connectors	5
	3. Jacks	5
	E. Types of Shielding Symbols Used	6
	1. Single-Conductor Shielded Wire	6
	2. Twisted-Pair Nonshielded Wire	6
	3. Twisted-Pair Shielded Wire	6
	4. Twisted-Triple Wires Grouped	7
	5. Non-Critical Shield	7
	6. Critical Shield	7
	7. Shield Symbol	7
	F. Position and limit Switch Codes	7
	G. Operator Switch Nomenclature and Contact Condition	8
II.	Elementary Diagram Symbols and Application Data	9
	A. Explanation of Digital Symbols	9
	1. Logic Type Code	9
	2. Element Type Code	9
	3. Functional Name or Mnemonic	10
	4. Internal Symbols	10
	5. Pin Numbers	10
	6. Location Terms	10
	B. Symbols with Application Data for Digital Cards	12
	C. Analog and Hybrid Symbols	19
III.	Tables	22
	A. Logic Code - Table 1	22
	B. Circuit Type Code - Table 2	22
	C. Circuit Class Code - Table 3	23
	D. Internal Symbol - Table 4	23

I. NOTES ON READING THE ELEMENTARY DIAGRAM

A. Information designating coil nomenclature, coil location code and contact locations is provided for each relay. Designation practices are illustrated in the following example.



Coil nomenclature and location code information appears adjacent to the relay symbol. The nomenclature defines the particular function the relay serves within the control system. A complete listing of these functions is provided on sheets A01 through A06 of the Turbine Control Elementary.

The coil location code defines the physical position of the relay within the control panel. In the example, 1ROCA is the location code and is made up as follows:

The first two characters define the module on which the device is mounted. In this case the relay is mounted on module 1R, the turbine control panel relay module.

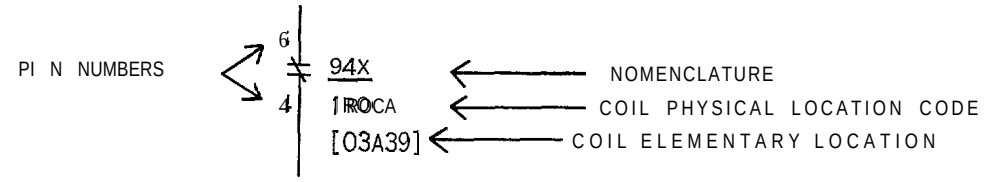
The third character designates the row number on the module. In this case the relay is mounted on row 0.

The fourth character defines the slot position in the row. In this case the relay is mounted in slot C of row 0.

If there is more than one position in a given slot, then a fifth character is used to define this position. There are three positions per slot in the relay panel, therefore, the last digit of this example, A, designates position A of slot C. (The positions are labeled A, B, C from top to bottom of slot .)

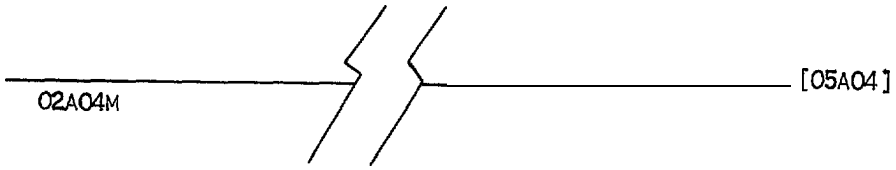
Contact location information shown in the margin below (or to the right of) the *relay* coil symbol designates the place within the elementary diagram where relay contacts appear. The first three characters of the code are the sheet number, and the last two digits define the horizontal line on which the contacts are shown, Normally closed contacts are designated by an underlined location number.

Information adjacent to each relay contact in the elementary diagram includes the nomenclature and coil location code, relay receptacle pin numbers, and coil elementary location. The following example illustrates this.



The nomenclature and coil location code are identical to the associated relay information. The location of the coil symbol in the elementary [in brackets] is similar to the contact locations given with the relay symbol. In this example, [03A39] indicates that the relay coil is shown on sheet 03A, horizontal line 39.

B. Wires entering or leaving an elementary sheet are referenced to their origin or destination, respectively. For example:



The location number under the incoming wire indicates where the wire originates. The sheet number, horizontal line, and vertical line are given. The number in brackets adjacent to the wire leaving the sheet gives the destination of the wire. The sheet number and horizontal line are given. In this case the wire entering the sheet comes from sheet 02A, horizontal line 04, and vertical line M and leaves the sheet going to sheet 05A, horizontal line 04.

C. Potentiometers shown in the elementary are located on the front of the card and can be adjusted with the card in place. The location of the potentiometer on the card is defined by the number in parenthesis beside the pot nomenclature, The pots are labeled in consecutive order beginning with 1 at the top and increasing toward the bottom of the card.

Example: R47(3) is the third potentiometer from the top of the card.

D. Terminal Boards and Connectors

1) Terminal boards are used in the turbine control panel to provide convenient termination points for wiring which must be used external to the panel. Conversely, all wiring from external sources to the turbine panel is also connected through terminal boards, except for thermocouple wires which are wired directly to the thermocouple averaging module or T. C. selector switch. High level and low level wiring are segregated by assigning

separate terminal boards to each. These are identified with the following nomenclature.

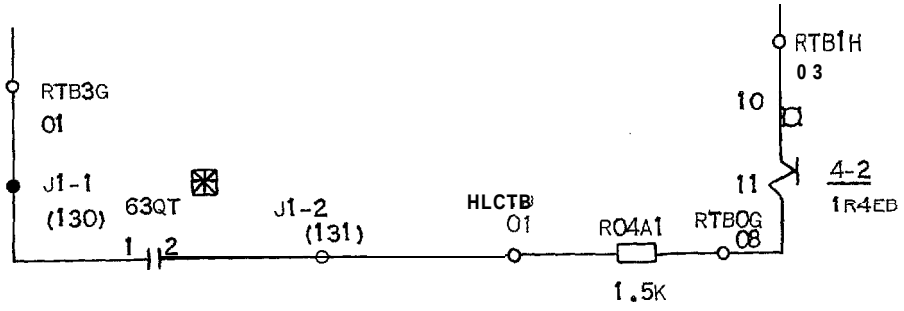
RTB - Relay Module Terminal Board (High Level)
LCTB1 - Low Level Customer Terminal Board
LCTB2 - Low Level Customer Terminal Board
HCTB1 - High Level Customer Terminal Board
ISOL - High Level Isolated and Filtered Customer Inputs

This nomenclature plus a TB designator (if required) and a terminal number appears above the symbol [——o—— 1 on the elementary diagram to indicate that the terminal board is used. (see the following examples)

2) Plug/receptacle type connectors (ribbon cable) are also shown by the symbol [——o—— I with the P/R designator and pin number appearing above the symbol. These connectors are used within the control panel for connections between subassemblies. (See examples for details)

3) Jacks (Pyle-National type) are used for connection between the packaged control cab and the turbine base. The symbol [——●——] with the jack designator and pin number appearing above the symbol is used in the elementary diagram to indicate where a jack is used. (See the following examples)

Examples:



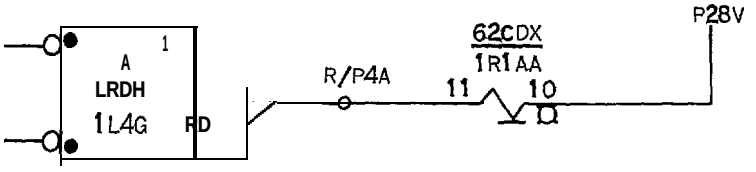
Example 1

Starting at the left of the diagram a wire within the control panel is connected to the relay terminal board on pin 1, row 3, and slot G. From there it leaves the control cab and goes to pin 1 of jack 1 in the turbine junction box. It leaves the junction box on wire (130) of an interconnecting cable to generator compartment to 63QT device. From the device it returns on wire (131) of the cable and connects to pin 2 of J1. From J1 it goes to the High Level Customer Terminal Board on the control panel. Resistor R04A is connected between HLCTB and relay terminal board row 0, slot G, pin 8. Relay 4-2 is connected between pin 8 of RTBOG and pin 3 of RTB1H. The wire leaves RTB1H pin 3 and goes to another subassembly within the panel.



Example 2

Relay driver LRDJ located in SPEEDTRONIC* page, 1L, is wired to light 1AIL on front door via "pigtail" (ribbon cable plug-D) from page 1L to receptacle RECP-D on front door. Then from RECP-D mounted on front door to light 1AIL by random wire.

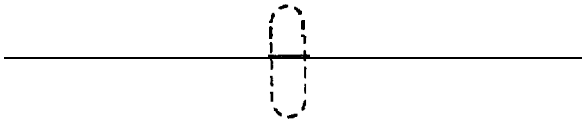


Example 3

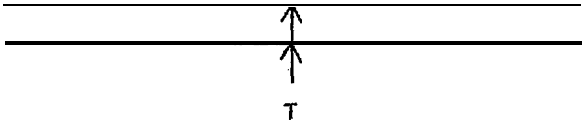
Relay driver LRDH located in SPEEDTRONIC* page, 1L is wired to receptacle RECP4A located on row 4, slot A of page. Ribbon cable plug PLUG4A is plugged into RECP4A and routed to relay module, 1R where it is "pigtail" wired to pin 11 of relay 62CDX located in row 1, slot A and position A.

E. Types of Shielding Symbols Used

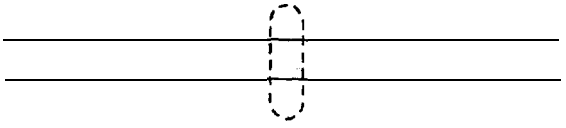
1) Single-Conductor Shielded Wire



2) Twisted-Pair Nonshielded Wire



3) Twisted-Pair Shielded Wire

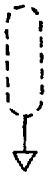


* Trademark of General Electric Co.

4) Where it is inconvenient to show twisted triple wires grouped together, a number is assigned following the twisted designator. In the following example, T1 indicates that each of the wires shown belong to twisted group 1.



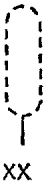
5) Non-Critical Shield



6) Critical Shield - The shield must be tied to the point specified by XX.



7) This symbol indicates the shield is a continuation of the shield tied to XX.



F. Position and limit switches have codes consisting of two lower case letters plus a number when required. The first letter indicates the contact condition corresponding to the end position of the device as indicated by the second letter. If a number is used, it indicates the percent of travel from the end position.

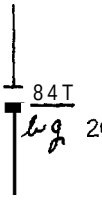
The contacts are shown in the position they return to normally when power is removed, or after a normal shutdown (if no position is

defined by this, the contacts will be shown normally open)

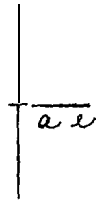
The following examples illustrate the use of the codes. A list of the codes is given following the examples.



OPEN ONLY WHEN STARTING CLUTCH (33CS DEVICE) IS FULLY DISENGAGED (END POSITION).



CLOSED ONLY WHEN GAS FUEL TRANSFER VALVE (84T DEVICE) IS 20% OR LESS FROM FULLY ON GAS (END POSITION).



OPEN ONLY WHEN STARTING CLUTCH IS FULLY ENGAGED.

First Letter: a = open
b = closed

Second Letter:

b = Bunker	e = Engaged	l = Lower
c = Closed	f = Forward	o = Open
d = Diesel	g = Gas	r = Raise
d = Disengaged	h = High	r = Retracted
d = Down	l = Liquid	t = Torque
		u = up

G. Operator switch nomenclature and contact condition is shown above the contacts on the elementary. The contacts are shown in the open condition. The position of the operator switch corresponding to closed contacts is shown between nomenclature and contact. The following example will illustrate this.

43
A.R.
6 II 6C

Contact 6 and 6C are made when the master operation selector switch, "43", is in auto or remote position.

II. ELEMENTARY DIAGRAM SYMBOLS AND APPLICATION DATA

A. Explanation of Digital Symbols

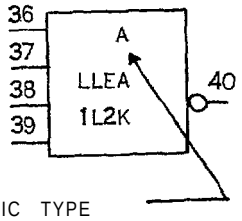
A logic element requires several identifications that are included on its symbols,

1. Logic type
2. Element type
3. Functional name
4. Internal symbols
5. Pin number
6. Physical location

1. Logic Type Code:

The logic type code identifies the internal function of the logic element, Refer to Table 1 for a list of logic type codes,

Example:

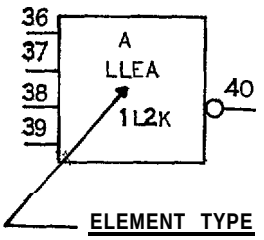


LOGIC TYPE
A = (AND)

2. Element Type Code:

The element type is shown on the logic diagram and is entered on the line below the logic type. The element type is always four characters which include circuit class (first letter), circuit type (second and third letter), and circuit: serial number (last letter). For circuit class refer to Table 3. For circuit type refer to Table 2. The circuit serial number is that term assigned to the circuit as time of initial design.

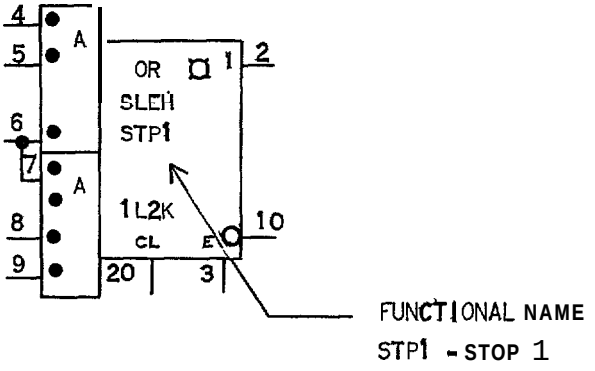
Example:



3. Functional Name or Mnemonic:

On some symbols the function that the element performs in the system is given a name which serves as a mnemonic to the reader. It is assigned by the designer and consists of a combination of alphabetical characters, numbers, or both.

Example:



4. Internal Symbols:

At times, it is required that internal functional information be shown to further identify the symbol. The example of the functional name previously shown will be used to illustrate this. Shown internally on this example are two separate and distinct inputs labeled E and CL. See Table 4 for a list of these terms.

5. Pin Numbers:

The pin number for every element or device is shown. This number is placed on or near the connection point and is located above horizontal lines or on the left of a vertical line.

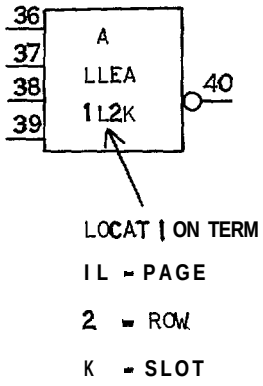
6. Location Terms:

Location terms locate a device physically and spatially but do not identify it. The location term is entered below the

functional name on the symbol, The location term has the following format:

X_1, X_2	X_3	X_4
Module, panel or page	Row	Slot

Example:



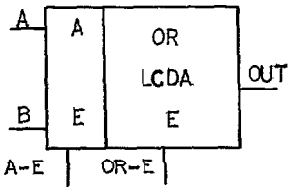
B. SYMBOLS WITH APPLICATION DATA FOR DIGITAL CARDS

NOTE: 1) AN IC3600L LOAD IS 1,715 MA. MAXIMUM

2) AN IC3600V LOAD IS 1.5MA OR 7/8 OF L LOAD.

LCDA

CLOCK/LOGIC DRIVER
9 CKTS/CARD



INPUTS
1 LOAD EACH

OUTPUTS
WILL DRIVE
32 LOADS EACH

POWER REQD (MA)
12V - 185 MAX.
115 MIN.

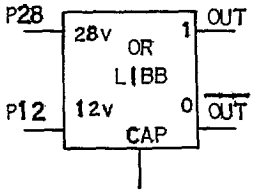
OUT = A . B IF EXPANDERS NOT USED

CIRCUIT	A	B	C	D	E	F	G	H	I
A	2	7	12	17	22	31	36	41	46
B	3	8	13	18	23	32	37	42	47
A-E	4	9	14	19	24	33	38	43	48
OR-E	6	11	16	21	26	34	39	44	49
OUT	5	10	15	20	25	35	40	45	50

IC3600LCDA

LIBB

INPUT BUFFER
9 CKTS/CARD



$OUT = P28 + P12$

$\overline{OUT} = \overline{P28} \cdot \overline{P12}$

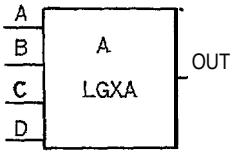
INPUTS
CONTACT CLOSURES TO P28V AND P12V

OUTPUTS
LOGIC 1 AND 0 EACH WILL DRIVE 5 LOADS

CIRCUIT	A	B	C	D	E	F	G	H
28V IN	2	7	12	17	22	31	36	41	46
12V IN	3	8	13	18	23	32	37	42	47
CAP	6	11	16	21	26	33	38	43	48
OUT	5	10	15	20	25	35	40	45	50
$\overline{OUT}$	4	9	14	19	24	34	39	44	49

IC3600LIBB1

LGXA



GATE EXPANDER
9 CKTS/CARD

OUT = A • B • C • D

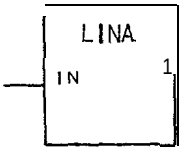
POWER REQD
NONE

OUTPUTS
USED AS INPUTS TO AND - EXPAND GATES OF OTHER ELEMENTS

CIRCUIT	1	2	3	4	5	6	7	8	9
A	2	7	12	17	22	31	36	41	46
B	3	8	13	18	23	32	37	42	47
C	4	9	14	19	24	33	38	43	48
D	6	11	16	21	26	34	39	44	49
OUT	5	10	15	20	25	35	40	45	50

IC3600LGXA1

LINA



LIGHT INDICATOR
16 CKTS/CARD

INPUTS
1 LOAD EACH

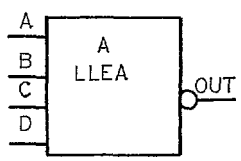
OUTPUTS
LIGHT INDICATION

POWER REQD (MA)
12V-225MAX
45 MIN

INPUT PIN NUMBERS
3,7,9,11,13,17,19,21,31,33,37,39,41,43,47,49

IC3600LINA1

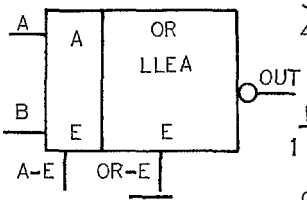
LLEA



OUT = $\overline{A \cdot B \cdot C \cdot D}$

POWER REQD (MA)

12V - 24 MAX
18 MIN



OUT = $\overline{A \cdot B}$ IF EXPANDER NOT USED

LOGIC ELEMENT
9 CKTS/CARD
5-2 INPUT CKTS
4-4 INPUT CKTS

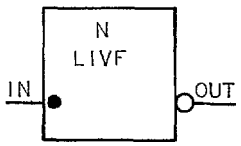
INPUTS
1 LOAD EACH

OUTPUTS
WILL DRIVE 10 LOADS EACH

CIRCUIT	1	2	3	4	5	6	7	8	9
A	2	7	12	17	22	31	36	41	46
B	3	8	13	18	23	32	37	42	47
C	—	—	—	—	—	33	38	43	48
D	—	—	—	—	—	34	39	44	49
A-E	4	9	14	19	24	—	—	—	—
OR-E	6	11	16	21	26	—	—	—	—
OUT	5	10	15	20	25	35	40	45	50

IC3600LLEA1

LIVF



OUT = $\overline{IN}$

POWER REQD (MA)

12V - 37.5 MAX
28 MIN

LOG I C I NVERTER
16 CKTS/CARD

INPUTS
1 LOAD EACH

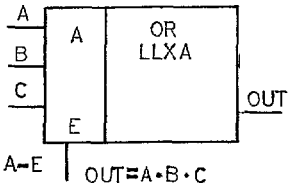
OUTPUTS
WILL DRIVE 5 LOADS EACH

CIRCUIT	1	2	3	4	5	6	7	8
IN	3	7	9	11	13	17	19	21
OUT	4	8	10	12	14	18	20	22

CIRCUIT	9	10	11	12	13	14	15	16
IN	31	33	37	39	41	43	47	49
OUT	32	34	38	40	42	44	48	50

IC3600LIVF

LLXA



POWER REQD (MA)

12V - 16 MAX
10 MIN

LOG I C EXPANDER
9 CKTS/CARD

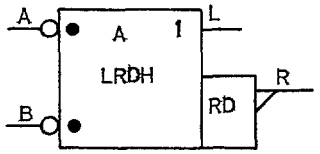
INPUTS
1 LOAD EACH

OUTPUTS
USED AS INPUTS TO OR-EXPAND GATES OF OTHER ELEMENTS

CIRCUITS	1	2	3	4	5	6	7	8	9
A	2	7	12	17	22	31	36	41	46
B	3	8	13	18	23	32	37	42	47
C	4	9	14	19	24	33	38	43	48
A-E	6	11	16	21	26	34	39	44	49
OUT	5	10	15	20	25	35	40	45	50

IC3600LLXA1

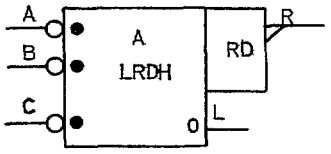
LRDH



CKT 1-5

$L = \bar{A} \cdot \bar{B}$

$R = \bar{A} \cdot \bar{B}$ (RELAY ENERGIZED)



CKT 6,7

$L = A' + B + C$

$R = \bar{A} \cdot \bar{B} \cdot \bar{C}$ (RELAY ENERG.)

NOTE: R PROVIDES CLOSURE TO COMMON TO ENERGIZE RELAY

CKT.	1	2	3	4	5	6	7
A	20	30	35	41	48	3	12
B	22	31	36	42	49	4	13
C	-	-	-	-	-	6	15
L	19	24	34	40	47	7	16
R	23	32	37	44	50	2	11

RELAY DRIVER

7 CKTS CARD

INPUTS

1 LOAD EACH

OUTPUTS

L - 5 LOADS EACH

R - 200 MA MAX.

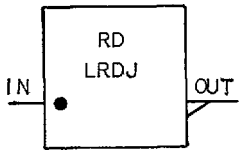
PWR REQU I RED

12V - 135 MA MAX

30 MA MIN

IC3600LRDH

LRDJ



OUT= IN

RELAY DRIVER

16 CKTS CARD

DIODE CLAMP PROVIDED

PWR REQU I RED

12V - 276 MA MAX

50 MA MIN

I NPUT

2 LOADS EACH

OUTPUT

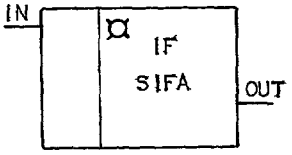
100 MA MAX.

NOTE : OUT PROVIDES CLOSURE TO COMMON TO ENERGIZE RELAY

CKT	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
IN	3	7	9	11	13	17	19	21	31	33	37	39	41	43	47	49
OUT	4	8	10	12	14	18	20	22	32	34	38	40	42	44	48	50

IC3600LRDJ

SIFA



P28 ON IN GIVES P12 ON OUT

INPUT FILTER
9 CKTS/CARD

INPUT
5.3 MA LOAD EACH

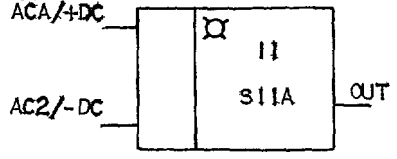
OUTPUT
2.5 MA DRIVE
FOR AN IC3600LIBB

PWR REQUIRED
NONE

CKT	1	2	3	4	5	6	7	8	9
IN	1	2	3	4	5	6	7	8	9
OUT	5	10	15	20	25	35	40	45	50
IND	IL1	IL2	IL3	IL4	IL5	IL6	IL7	IL8	IL9

IC3606SIFA9

SIIA



ISOLATED INPUTS
9 CKTS/CARD

INPUT
95-120 VDC
97-132 VAC (RMS)
3.5MA LOAD EACH

OUTPUT
2.5MA DRIVE FOR A IC3600LIBBA

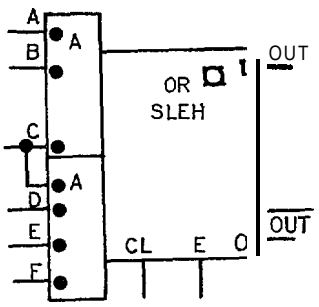
OUT = 11VDC WHEN INPUT
IS 95-120 VDC CR 97-132 VAC

PWR REQUIRED
12V ⎓
29MA MAX
25MA MIN

CIRCUIT	1	2	3	4	5	6	7	8	9
AC1/+DC TERMINAL	1	2	3	4	5	6	7	8	9
AC2/-DC TERMINAL	11	12	13	14	15	16	17	18	19
OUT PIN #	5	10	15	20	25	35	40	45	50
INDICATING LIGHT	IL1	IL2	IL3	IL4	IL5	IL6	IL7	IL8	IL9

IC3606SIIA9

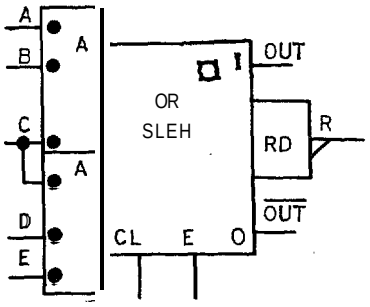
SLEH



$OUT = (A \cdot B \cdot C) + (C \cdot D \cdot E \cdot F)$

$\overline{OUT} = \overline{C} + (\overline{A} + \overline{B}) \cdot (\overline{D} + \overline{E} + \overline{F})$

CKTS 1 AND 2



$R = (A \cdot B \cdot C) + (C \cdot D \cdot E)$

CKTS 3, 4, AND 5

LOGIC ELEMENT

5 CKTS/CARD

PWR. REQ'D

12V - 150 MA MAX

25 MA MIN

OUTPUTS

1 AND 0 - CAN DRIVE 12 LOAD EACH

R' - 350 MA MAX

INPUTS

1 LOAD EACH

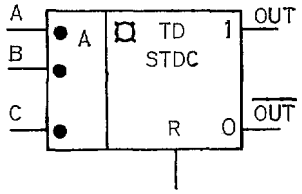
E-5 LX GATES

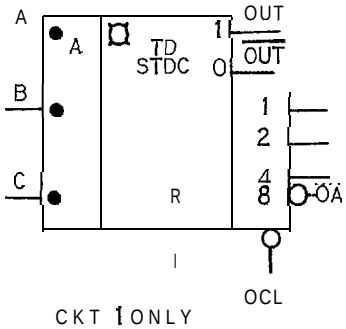
CL-10 LOADS

CKT	A	B	C	D	E	F	OR-E	OUT	$\overline{OUT}$	R	CL
1	4	5	6	7	8	9	3	2	10	-	20
2	13	14	15	16	17	18	12	11	19	-	20
3	24	28	29	30	31	-	25	23	32	21	20
4	35	37	38	39	40	-	36	34	41	33	20
5	44	46	47	48	49	-	45	43	50	42	20

IC3600SLEH

STDC





CKT 1 ONLY

TIME DELAY

4 CKTS/CARD

PWR. REQ' D	
P12 =	23 MA MAX
	17 MA MIN
N12 =	13 MA MAX
	9 MA MIN

INPUTS A, B, AND C = 2 LOADS EACH
OCL = 4V LOADS
R = RANGE ADJUST = WITH R OPEN, TIME DELAY RANGE IS 1 SEC TO 30 SEC.
WITH R TIED TO N12, THE RANGE IS 10 TO 300 SEC.

OUTPUTS EACH CAN DRIVE 5 20KHZ LOADS
OUT- BECOMES A "1" AFTER (A·B·C) HAS BEEN "1" FOR THE SET TIME DELAY.
$\overline{\text{OUT}}$- BECOMES A "1" IMMEDIATELY AFTER (A·B·C) BECOMES "0".

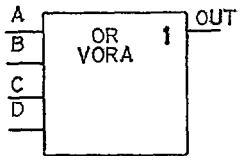
NOTE 1. IF THE 0 OUTPUT IS CONNECTED TO ONE OF THE INPUTS, THE CIRCUIT WILL BECOME A FREE RUNNING OSC AS LONG AS THE OTHER INPUTS ARE PERMISSIVE. THIS MODE OF OPERATION IN CONJUNCT1 ON WITH THE COUNTER ELEMENTS IS USEFUL IN VARIABLE DELAYS LONGER THAN 5 MINUTES.

NOTE 2. A DIVIDER IS PROVIDED ON CIRCUIT 1 WHICH WILL OUTPUT A "0" ON "OA" AFTER THE TIMER HAS CYCLED 8 TIMES. NORMALLY OA IS TIED BACK TO ONE OF THE INPUTS TO INHIBIT FURTHER CYCLING AFTER EIGHT CYCLES.

CKT	A	B	C	OUT	$\overline{\text{OUT}}$	OA	OCL	R
1	2	3	4	7	11	24	22	9
2	13	14	15	17	18	-	-	20
3	31	32	33	37	39	-	-	38
4	42	43	45	48	50	-	-	49

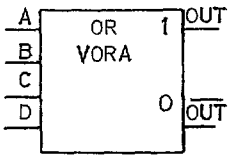
IC3600STDC

VORA



CKT 1 AND 2

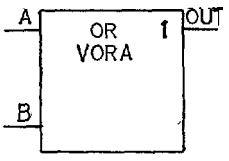
OUT= A+B+C+D



CKT 3 AND 4

OUT=A+B+C+D

$\overline{OUT}$=A+B+C+D



CKT 5-12

OUT=A+B

OR LOGIC

12 CKTS/CARD

INPUTS

1V LOAD EACH

OUTPUTS

CAN DRIVE 8V LOADS

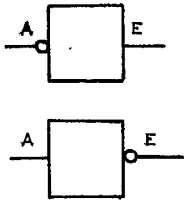
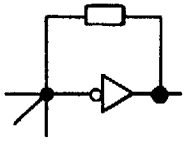
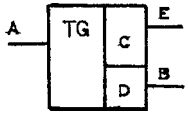
PWR. REQ'D

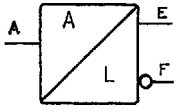
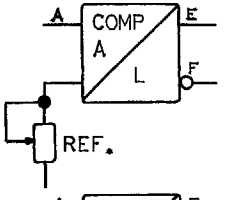
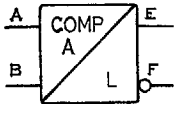
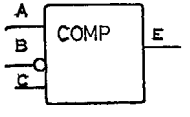
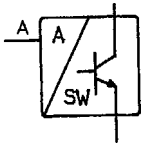
P5.3 = 90 MA MAX

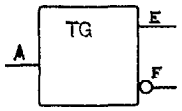
65 MA MIN

CKT	1	2	3	4	5	6	7	8	9	10	11	12
A	6	10	16	23	30	32	35	33	46	44	50	48
B	5	9	14	22	36	26	34	31	45	43	49	47
C	4	7	13	21	-	-	-	-	-	-	-	-
D	3	8	12	20	-	-	-	-	-	-	-	-
OUT	15	11	2	18	25	24	39	38	37	40	42	41
$\overline{OUT}$	-	-	17	19	-	-	-	-	-	-	-	-

IC3600VORA

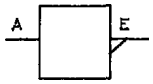
SYMBOL	DESCRIPTION
1 0	REPRESENTS THE TWO STATES OF BINARY LOGIC 1 = ENERGIZED (A POSITIVE VOLTAGE) 0 = DE-ENERGIZED (A NEAR ZERO VOLTAGE)
	**INVERSION** A CIRCLE AT THE INPUT OR OUTPUT OF A SYMBOL REPRESENTS A SIGNAL INVERSION A = 0 → E = 1 A = 1 → E = 0
	**OPERATIONAL AMPLIFIER WITH NEGATIVE FEEDBACK** THE SUMMING JUNCTION IS REPRESENTED BY A CIRCLE. THE INVERSION SYMBOL ON THE AMPLIFIER INDICATES A NEGATIVE OUTPUT FOR A POSITIVE INPUT.
	**TRIGGER, ANALOG TO LOGIC** A = ANALOG > DEAD BAND RANGE → E = 1, B = 0 A = ANALOG < DEAD BAND RANGE → E = 0, B = 1 A = ANALOG WITHIN DEAD BAND RANGE E = 0, B = 0

SYMBOL	DESCRIPTION
	**ANALOG TO LOGIC** A = ANALOG > REFERENCE → E = 1, F = 0 A = ANALOG < REFERENCE → E = 0, F = 1
	**ANALOG REFERENCE COMPARISON TO LOGIC** A = ANALOG > REFERENCE → E = 1, F = 0 A = ANALOG < REFERENCE → E = 0, F = 1
	**ANALOG DIFFERENTIAL COMPARISON TO LOGIC** ANALOG A = ANALOG B → E = 0, F = 1 ANALOG A > ANALOG B → E = 1, F = 0 ANALOG A < ANALOG B → E = 1, F = 0
	**ANALOG DIFFERENTIAL COMPARISON TO LOGIC, GATED** B = 1 E = 0 B = 0 AND ANALOG A = ANALOG C → E = 0 B = 0 AND ANALOG A > ANALOG C → E = 1 B = 0 AND ANALOG A < ANALOG C → E = 1
	**LOGIC SWITCH** A = 0 → TRANSISTOR OFF A = 1 → TRANSISTOR ON



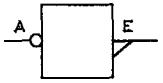
TRIGGER, ANALOG TO LOGIC

A = ANALOG > REFERENCE E = 1, F=0
A = ANALOG < REFERENCE E = 0, F=1

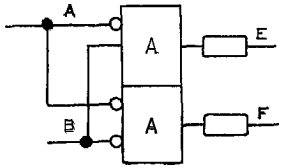


RELAY/LIGHT DRIVER

A=1 OR QUASI 1 (IE. A HIGHER VOLTAGE AS CONTRASTED TO A LOWER VOLTAGE) CAUSES E TO SWITCH TO COMMON TO ENERGIZE A LIGHT OR RELAY.

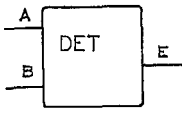


(INVERSE OF ABOVE)



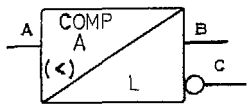
ANALOG SELECTION

WHEN A = 0 AND B = 1 → E = ANALOG OF " NORM " MAGNITUDE
WHEN A = 0 AND B = → 0 F = ANALOG OF " FAST "



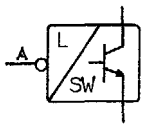
DETECTOR

A.C. ON A, B RECTIFIED TO DC ON E AND COMM.



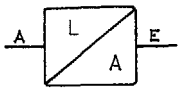
COMPARATOR

IF A IS LESS THAN SOME PRESET VALVE, B = 1 AND C = 0
OTHERWISE B = 0, C = 1



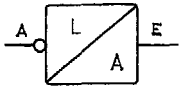
LOGIC SWITCH

A = 0-TRANSISTOR ON
A = 1--TRANSISTOR OFF



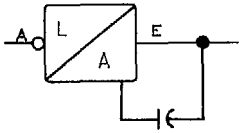
LOGIC TO ANALOG

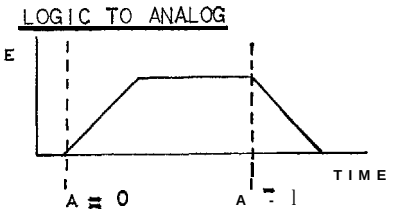
A = 0 - - - E = 0
A = 1 → E = FIXED ANALOG

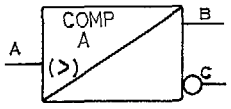


LOGIC To ANALOG

A = 0 → E = FIXED ANALOG
A = 1 → E = 0

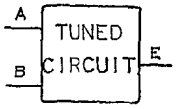






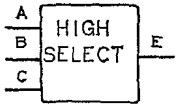
COMPARATOR

IF A IS GREATER THAN SOME PRESET VALVE, B = 1, C = 0
OTHERWISE B = 0, C = 1



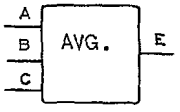
TUNED CIRCUIT

A, B VARIABLE FREQUENCY $\geq$ REFERENCE → E = 1
AND ALSO ENERGIZES A RELAY



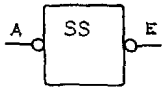
HIGH SELECT

ANALOG E = HIGHEST OF ANALOGS A, B, C

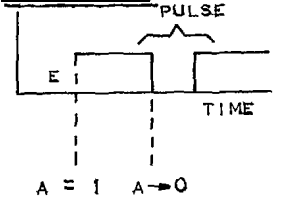


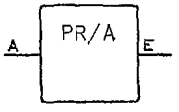
AVERAGE

ANALOG E = AVERAGE OF ANALOGS A, B, C



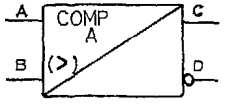
SINGLE SHOT





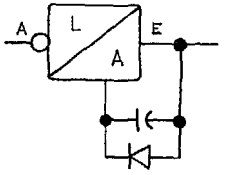
PULSE RATE TO ANALOG

A = D.C. PULSES OR A.C. → E = ANALOG D.C.
PROPORTIONAL TO FREQUENCY OF A

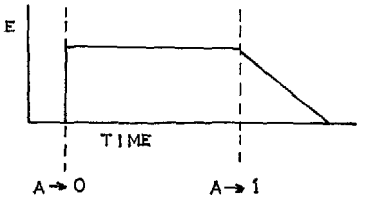


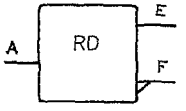
COMPARATOR

IF A IS GREATER THAN B
THEN C = 1 , D = 0
OTHERWISE C = 0 , D = 1



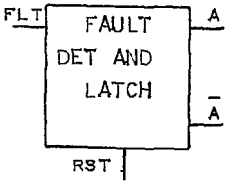
LOGIC TO ANALOG





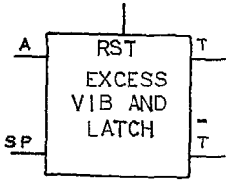
DRIVER

A = 0 → E = 0, F = RELAY/LIGHT DE-ENERGIZED
A = 1 → E = 1, F = RELAY/LIGHT ENERGIZED



FAULT DETECTOR AND LATCH

SHORT OR OPEN ON FLT
SETS A = 1, $\overline{A}$ = 0
1 ON RST RESETS LATCH



VIBRATION DETECTOR AND LATCH

A $\geq$ SP → T = 1, $\overline{T}$ = 0
RST = 1 → RESETS LATCH

TABLE 1 LOGIC CODES

CODES	LOGIC TYPE	CODE	LOGIC TYPES
A	AND	SS	SINGLE SHOT
DA	DIGITAL AMPLIFIER	SW	SHIFT WINDING
FF	FLIP-FLOP	TD	TIME DELAY
I	INDICATOR	OS	OVERSPEED SENSOR
IF	INPUT FILTER		
II	ISOLATED INPUT		
N	LOGIC INVERTER		
OE	EXCLUSIVE OR		
OR	OR		
osc	OSCILLATOR		
RD	RELAY DRIVER		
RC	REVERSING COUNTER		
SR	SHIFT REGISTER		

TABLE 2 CIRCUIT TYPE CODE

CODE	TYPE	CODE	TYPE
CD	CLOCK/LOGIC DRIVER	PD	PEAKING OPERATION
FP	GENERATOR DRIVE	RC	REVERSIBLE COUNTER
FK	FUEL SPLITTER	RD	RELAY DRIVER
FU	FUEL FLOW	SL	SETPOINT CONTROL
FI	FLAME INDICATOR	SK	START-UP CONTROL
GX	GATE-EXPANDER	sz	SPEED CONTROL
GD	GROUND DETECTOR	sv	SERVOVALVE DRIVE
HP	HIGH PRESSURE	TD	TIME DELAY
IV	INVERTER VOLTAGE	TK	TEMPERATURE CONTROL
LE	LOGIC ELEMENT	VD	VIBRATION DETECTOR
LX	LOGIC EXPANDER	vs	SPEED SENSOR
OR	"OR" LOGIC		
ox	OVERSPEED SENSOR		
OT	OVERTEMPERATURE		
OS	OSCILLATOR		

TABLE 3 CIRCUIT CLASS CODE

CODE	CIRCUIT CLASS
A	ANALOG
H	HIGH-SPEED LOGIC (300KC AND UP)
M	MEDIUM SPEED LOG I C (20KC TO 500KC)
L	LOW SPEED LOG I C (DC TO 20KC)
P	PER I PHERAL SPEC I FI CALLY REQUI RED FOR PER I PHERAL DEV I CE
S	SPECI AL (ANY CLASS NOT SPECI FI CALLY DEFINED)
K	RELAYS ONLY (NO LOGIC)
V	BASIC STATIC LOGIC

TABLE 4 INTERNAL SYMBOL

SYMBOL	MEANING	SYMBOL	MEANING
CL	CLEAR	INH	INHIBIT
CAP	CAPACITOR TIE	•	HIGH LEVEL INPUT*
COM	COMMON	G	GATE
C	CARRY	LD	LOAD
IN	I NPUT		
OUT	OUTPUT		
S	SET		
R	RANGE		
E	EXPANDER		
PUL	PULSE		
STO	ENABLE UP COUNT		
ST1	ENABLE DOWN COUNT		
C L	INHIBIT CLEAR		
OCL	CLEAR ALL BITS		
S	SET		
RST	RESET		

* LOGIC "0" ≤ 1.5 VOLTS
LOGIC "1" > 4.75 VOLTS

WHICH MEANS THAT EXPANDER GATE LGXA CAN BE USED

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL@ ELECTRIC